Code No: 3411

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) IV-Semester (MAIN) Examinations July-2025

DATA ANALYTICS USING R

Time : 3 hours Max. Marks : 60

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What is the purpose of the function "sort()" in R programming Language? Write the code to arrange the following in i) decreasing and ii) ascending order and for printing the output. Data: 1,3, 0, 29, 5, 7, 10, 9, 12	2	2	1
b)	What is data frame in R Program? Write a R program for creating a data frame for fruits with the following sub fields of fruits • Colours • Type • Weight • Price	2	3	1
c)	What is Logistic regression, and write its significance in Machine Learning? Give an example. Identify the type of classification for the Fig 1 and Fig 2	2	5	2

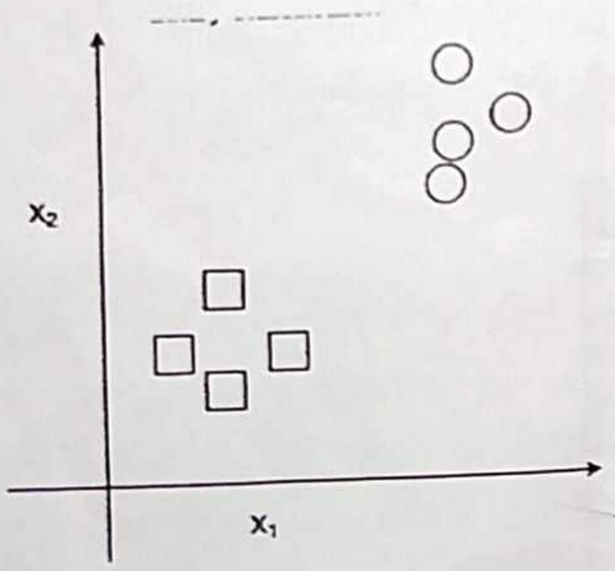


Fig 1

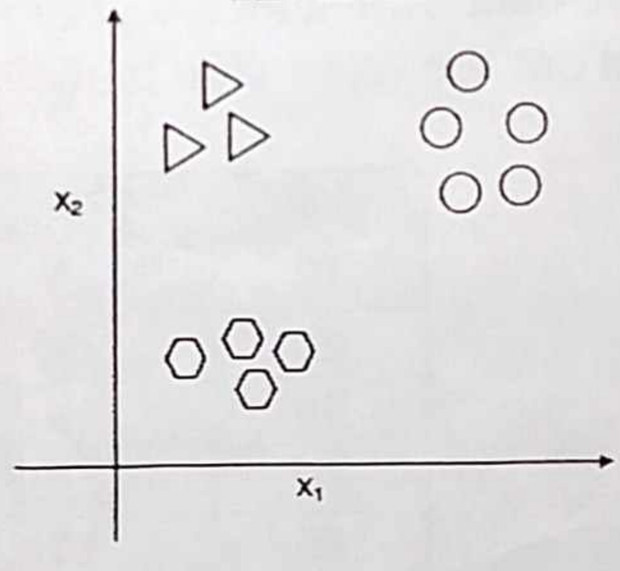


Fig 2

d)	What is clustering? List out the algorithms used in Clustering.	2	1	2
e)	What is text Mining? Briefly explain what are the steps in carrying out the Text mining for a given " text data".	2	2	4

(P.T.O.)

f) Following is the R program for Time Series data. What will be the output after running the following program. Please write down the output: 2

```
install.packages("forecast")
install.packages("tseries")
library(forecast)
library(tseries)
sales_data <- c(200, 220, 210, 230, 250, 270, 300, 320, 310, 330,
350, 370, 210, 230, 220, 240, 260, 280, 310, 330, 320, 340, 360,
380)
sales_ts <- ts(sales_data, start = c(2023, 1), frequency = 12)
print(sales_ts)
```

2. a)	List out what are "R " language functions for reading the data from various sources like file, table, keyboard etc. Write the syntax of the same along with an example.	4	1	1
b)	Write a program in R to represent the following data in a matrix as per below instructions: i) Matrix in row wise and print the output Data: 10, 2, 24, 17, 9, 11, 15, 19, 25, ii) Matrix having 3 columns and the print Data: 1,2, 3,4,5,6,7,8,9,10, 12 iii) Calculate the sum of the above two matrices and print result in a matrix	8	3	1
3. a)	i) What is Decision Tree in Machine learning, and write its significance? List out 3 Decision tree algorithms used in Machine Learning ii) Write the flowchart for ID3 Algorithm	5	2	2
b)	Following is the data of a selected group of people. Please write a R program to find out the regression model and also to print the graph.	7	4	2

Age	Weight
10	40
20	50
30	60
40	70
80	110

4. a)	What is K means algorithm? Write down the pseudo code? What is its application in Data Analysis in developing the AI/ML model?	5	2	4

b) We have 4 medicines as our training data points object and each medicine has 2 attributes.

Each attribute represents coordinate of the object. The object being to know which medicines belong to which cluster.

Calculate the distance matrix for **2 iterations** only for the following given data.

Object	Attribute1 (X): weight index	Attribute2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

7 5 4

5. a) What is Text Mining and explain the text mining process? Discuss the following Text mining methodologies. Give suitable examples

i) Sentiment Analysis
ii) Entity Recognition

7 3 3

b) Following is the R program for Parallel computing in R. Please list out what will be the output of the following R program:

```
library(parallel)
numbers <- list(1, 2, 3, 4, 5)
square <- function(x) {
Sys.sleep(1) # simulate some delay
return(x^2)
}
result <- mclapply(numbers, square, mc.cores = 2)
print(result)
```

5 6 3

6. a) i) In Time Series analysis using R, define what is "Forecasts using Exponential smoothing"?

ii) Explain the following smoothing **HoltWinters()** function in R and the syntax in R

6 3 4

b) What is ARIMA model in Forecasting?
What are the steps involved in using the Model in R.
List the syntax of **arima()** function in R

6 4 4

7. a) Define the following data structures in R with example code.
i) Data frame
ii) Vector
iii) Matrix

5 1 1

Code No. 7 5 3

b) Calculate the **Confidence** of the following association rule for the given following purchase data
{Bread, Butter} => {Milk}

TID	Items
1	Bread, Butter, Milk
2	Bread, Butter
3	Beer, Cookies, Diapers
4	Milk, Diapers, Bread, Butter

****